

Draft Guidance on Semaglutide

October 2025

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Active Ingredient: Semaglutide

Dosage Form: Tablet

Route: Oral

Strengths: There are two formulations (i.e., formulation R1 and formulation R2) with different recommended dosages. These formulations are not substitutable on a mg per mg basis. Strengths approved for each formulation include the following:

Formulation R1: 3 mg, 7 mg, 14 mg

Formulation R2: 1.5 mg, 4 mg, 9 mg

Recommended Studies: Demonstrate active pharmaceutical ingredient (API) sameness, comparative assessment of impurity, and two options to demonstrate bioequivalence: (1) six in vivo bioequivalence studies with pharmacokinetic endpoints or (2) two in vivo bioequivalence studies with pharmacokinetic endpoints and in vitro testing

Recommendations for demonstrating API sameness and comparative impurity assessment

Semaglutide can be produced using synthetic or semi-synthetic recombinant deoxyribonucleic acid (rDNA) methods. Provide sufficient data and justification to support API sameness (e.g., same primary sequence and physiochemical properties) and compare API-related impurity profile differences between the test product and reference listed drug (RLD).

Recommendation for demonstrating bioequivalence:

I. Option I: Six in vivo bioequivalence studies with pharmacokinetic endpoints

1. Type of study: Fasting
Design: Single-dose, two-treatment, two-period crossover in vivo
Strength: 14 mg
Subjects: Healthy males and non-pregnant, non-lactating females
2. Type of study: Fasting
Design: Single-dose, two-treatment, two-period crossover in vivo
Strength: 9 mg
Subjects: Healthy males and non-pregnant, non-lactating females
3. Type of study: Fasting
Design: Single-dose, two-treatment, two-period crossover in vivo
Strength: 7 mg
Subjects: Healthy males and non-pregnant, non-lactating females
4. Type of study: Fasting
Design: Single-dose, two-treatment, two-period crossover in vivo
Strength: 4 mg
Subjects: Healthy males and non-pregnant, non-lactating females
5. Type of study: Fasting
Design: Multiple-dose (e.g., administer single 3 mg dose for 5 days), two-treatment, crossover in vivo
Strength: 3 mg
Subjects: Healthy males and non-pregnant, non-lactating females
6. Type of study: Fasting
Design: Multiple-dose (e.g., administer single 1.5 mg dose for 5 days), two-treatment, crossover in vivo
Strength: 1.5 mg
Subjects: Healthy males and non-pregnant, non-lactating females

II. Option II: Two in vivo bioequivalence studies with pharmacokinetic endpoints and in vitro testing

Semaglutide is co-formulated with salcaprozate sodium which facilitates the absorption of semaglutide after oral administration. Therefore, Option II is acceptable if the test product is qualitatively the same and quantitatively similar to the corresponding strengths of the RLD. A test product of oral semaglutide tablet is considered quantitatively similar if the change in the amount of salcaprozate sodium is within $\pm 10\%$ of the amount of salcaprozate sodium present in the RLD. In addition, the cumulative difference of all excipients, including salcaprozate sodium and expressed as a percentage (w/w) of total test tablet weight, in test product compared the corresponding strengths of the RLD is within $\pm 10\%$ (w/w). Changes to non-absorption enhancing excipients should not change their functionality in the dosage form and property of the dosage form.

1. Type of study: Fasting
Design: Single-dose, two-treatment, two-period crossover in vivo
Strength: 14 mg
Subjects: Healthy males and non-pregnant, non-lactating females
2. Type of study: Fasting
Design: Single-dose, two-treatment, two-period crossover in vivo
Strength: 9 mg
Subjects: Healthy males and non-pregnant, non-lactating females

Waiver request of in vivo testing: (a) 3 mg and 7 mg strengths based on (i) an acceptable bioequivalence study on the 14 mg strength, (ii) acceptable in vitro semaglutide dissolution testing of 3 mg, 7 mg, 14 mg strengths, and (iii) acceptable in vitro salcaprozate sodium dissolution testing of 3 mg, 7 mg, 14 mg strengths; (b) 1.5 mg and 4 mg strengths based on (i) an acceptable bioequivalence study on the 9 mg strength, (ii) acceptable in vitro semaglutide dissolution testing of 1.5 mg, 4 mg, 9 mg strengths, and (iii) acceptable in vitro salcaprozate sodium dissolution testing of 1.5 mg, 4 mg, 9 mg strengths

Additional comments:

- Semaglutide tablet should be administered following the administration instructions for the RLD.
- Ensure an adequate washout period between treatments in the crossover study due to the long elimination half-life of semaglutide. Alternatively, a parallel study design may be considered.
- If it is not feasible to achieve sufficient bioanalytical sensitivity to adequately characterize the pharmacokinetic profile of 1.5 or 3 mg strength even after multiple doses in the fasting pharmacokinetic study, the applicant may submit a pre-abbreviated new drug application (ANDA) meeting request to discuss alternative bioequivalence approach for the 1.5 or 3 mg strength. The proposed alternative bioequivalence approach should be scientifically justified and satisfy the requirements of the applicable statutes and regulations.

- Monitor blood glucose concentrations and signs and symptoms of hypoglycemia during the study. Implement appropriate hypoglycemia management protocol.
- Due to the potential impact of semaglutide-associated gastrointestinal adverse events on subject dropout rates and study power, incorporation of safety measures in study design may be considered. The applicant should provide justification that these measures do not confound the study results.
- A replicate crossover study design (partial or fully replicate) is acceptable whether the reference product is a highly variable drug or not. However, if the plan is to use the reference-scaled average bioequivalence approach for bioequivalence study data analysis, provide evidence of high variability in the pharmacokinetic parameters (i.e., within-subject variability $\geq 30\%$) of the RLD. For detailed information on this approach, refer to the most recent version of the FDA guidance for industry on *Bioequivalence Studies with Pharmacokinetic Endpoints for Drugs Submitted Under an Abbreviated New Drug Application*.^a

Analyte to measure: Semaglutide in plasma

Bioequivalence based on (90% CI): Semaglutide

Dissolution test method and sampling times: The dissolution information for this drug product can be found in the FDA's Dissolution Methods database, <http://www.accessdata.fda.gov/scripts/cder/dissolution/>. Conduct comparative dissolution testing on 12 dosage units for each of all strengths of the test product and RLD.¹ Specifications will be determined upon evaluation of the ANDA.

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^a For the most recent version of a guidance, check the FDA guidance web page at <https://www.fda.gov/regulatory-information/search-fda-guidance-documents>.

¹ If the RLD is not available, refer to the most recent version of the FDA guidance for industry on *Referencing Approved Drug Products in ANDA Submissions*.